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In 1980, well before computers became household objects, the American physicist Paul Benioff demonstrated that a computer could theoretically operate under the laws of quantum physics. Today, there are many models of quantum computing. Quantum computers store and manipulate quantum information to perform calculations, potentially carrying out new calculations and algorithms that would offer an exponential time improvement on today's classical computers. Although serious technical barriers stand in the way of quantum computers entering the mainstream, they are expected to transform computing, communication, and security in the 21st century.